

DCL: it is known as data control language. It is a set of SQL component through which we control data. Like GRANT, REVOKE, COMMIT, ROLLBACK, SAVEPOINT

GRANT: Grant is a DCL component through which we give specific permission to guest user.

>mysql> grant select, insert, update on student to g1;

```
Microsoft Windows [Version 10.0.19045.6216]
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C:\Users\NandKishor>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.39 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| college |
| commerce |
| dsf |
| information_schema |
| mysql |
| performance_schema |
| sfspdb |
| sys |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> use dsf;
Database changed
mysql> show tables;
+-----+
| Tables_in_dsf |
+-----+
| student       |
| teacher       |
+-----+
2 rows in set (0.00 sec)

mysql> create user g1 identified by '123456';
ERROR 1396 (HY000): Operation CREATE USER failed for 'g1'@'%'
mysql> drop user g1;
Query OK, 0 rows affected (0.20 sec)

mysql> create user g1 identified by '123456';
Query OK, 0 rows affected (0.07 sec)

mysql> select user, host from mysql.user;
+-----+-----+
| user          | host          |
+-----+-----+
| g1            | %             |
| mysql.infoschema | localhost     |
| mysql.session  | localhost     |
| mysql.sys      | localhost     |
| root          | localhost     |
+-----+-----+
5 rows in set (0.00 sec)

mysql> grant select, insert, update on student to g1;
Query OK, 0 rows affected (0.06 sec)
```

```
Command Prompt - mysql -u g1 -p
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NandKishor>mysql -u g1 -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.39 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show grants;
+-----+
| Grants for g1@% |
+-----+
| GRANT USAGE ON *.* TO `g1`@`%` |
| GRANT SELECT, INSERT, UPDATE ON `dsf`.`student` TO `g1`@`%` |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> show databases;
+-----+
| Database |
+-----+
| dsf      |
| information_schema |
| performance_schema |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> use dsf;
Database changed
mysql> show tables;
+-----+
| Tables_in_dsf |
+-----+
| student       |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1001        | 1001        | Amit    | Indore  | DBMS    | 80    |
| DS1002        | 1002        | Sumit   | Bhopal  | Python  | 70    |
| DS1003        | 1003        | vinit   | Dhar    | DBMS    | 90    |
| DS1004        | 1004        | Deepak  | Ujjan   | OR      | 60    |
| DS1005        | 1005        | Ajay    | Indore  | DBMS    | 80    |
| Ds1006        | 1006        | Raj     | Indore  | Python  | 95    |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> update student set marks='98' where std_enroll_no='DS1006';
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1001        | 1001        | Amit    | Indore  | DBMS    | 80    |
| DS1002        | 1002        | Sumit   | Bhopal  | Python  | 70    |
| DS1003        | 1003        | vinit   | Dhar    | DBMS    | 90    |
| DS1004        | 1004        | Deepak  | Ujjan   | OR      | 60    |
| DS1005        | 1005        | Ajay    | Indore  | DBMS    | 80    |
| Ds1006        | 1006        | Raj     | Indore  | Python  | 98    |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> delete from student;
ERROR 1142 (42000): DELETE command denied to user 'g1'@'localhost' for table 'student'
mysql> _
```

REVOKE: Revoke is DCL component **through which we taking specific permission from guest user.**

Mysql> revoke select, insert, update on student from g1;

```
mysql> revoke select, insert, update on student from g1;
Query OK, 0 rows affected (0.04 sec)
```

Command Prompt - mysql -u g1 -p

Microsoft Windows [Version 10.0.19045.6216]

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C:\Users\NandKishor>mysql -u g1 -p

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 14

Server version: 8.0.39 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;

```
+-----+
| Database |
+-----+
```

```
| information_schema |
| performance_schema |
+-----+
```

```
| information_schema |
| performance_schema |
+-----+
```

```
| information_schema |
| performance_schema |
+-----+
```

```
| information_schema |
| performance_schema |
+-----+
```

```
| information_schema |
| performance_schema |
+-----+
```

2 rows in set (0.00 sec)

mysql> _

- **Mysql> Create user g1 identified by "123456;**
- **Mysql> select user, host from mysql.users;**
- **Mysql> Grant INSERT, UPDATE on student to g1;**
- **Login g1 (open new mysql command prompt)**
- **Show grants;**

3. Rollback : Rollback is a DCL component that are used to access previous original copy of database.

Mysql> start transaction;

Command Prompt - mysql -u root -p

```
mysql>
mysql> start transaction;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1001        | 1001        | Amit    | Indore  | DBMS    | 80    |
| DS1002        | 1002        | Sumit   | Bhopal  | Python  | 70    |
| DS1003        | 1003        | vinit   | Dhar    | DBMS    | 90    |
| DS1004        | 1004        | Deepak  | Ujjan   | OR      | 60    |
| DS1005        | 1005        | Ajay    | Indore  | DBMS    | 80    |
| Ds1006        | 1006        | Raj     | Indore  | Python  | 98    |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> delete from student where std_enroll_no='DS1004';
Query OK, 1 row affected (0.03 sec)

mysql> delete from student where std_enroll_no='DS1005';
Query OK, 1 row affected (0.00 sec)

mysql> delete from student where std_enroll_no='DS1006';
Query OK, 1 row affected (0.00 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1001        | 1001        | Amit    | Indore  | DBMS    | 80    |
| DS1002        | 1002        | Sumit   | Bhopal  | Python  | 70    |
| DS1003        | 1003        | vinit   | Dhar    | DBMS    | 90    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> rollback;
Query OK, 0 rows affected (0.06 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1001        | 1001        | Amit     | Indore  | DBMS     | 80     |
| DS1002        | 1002        | Sumit    | Bhopal  | Python   | 70     |
| DS1003        | 1003        | vinit    | Dhar    | DBMS     | 90     |
| DS1004        | 1004        | Deepak   | Ujjan   | OR        | 60     |
| DS1005        | 1005        | Ajay     | Indore  | DBMS     | 80     |
| Ds1006        | 1006        | Raj      | Indore  | Python   | 98     |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> _
```

4. Commit: to permanently store data in databases.

```
mysql> delete from student where std_enroll_no='DS1001';
Query OK, 1 row affected (0.07 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1002        | 1002        | Sumit    | Bhopal  | Python   | 70     |
| DS1003        | 1003        | vinit    | Dhar    | DBMS     | 90     |
| DS1004        | 1004        | Deepak   | Ujjan   | OR        | 60     |
| DS1005        | 1005        | Ajay     | Indore  | DBMS     | 80     |
| DS1006        | 1006        | Raj      | Indore  | Python   | 98     |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> rollback;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1002        | 1002        | Sumit    | Bhopal  | Python   | 70     |
| DS1003        | 1003        | vinit    | Dhar    | DBMS     | 90     |
| DS1004        | 1004        | Deepak   | Ujjan   | OR        | 60     |
| DS1005        | 1005        | Ajay     | Indore  | DBMS     | 80     |
| DS1006        | 1006        | Raj      | Indore  | Python   | 98     |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

5. Save point: To taking database backup at fix point

```
mysql> select * from student;
```

std_enroll_no	std_roll_no	std_name	address	subject	marks
DS1002	1002	Sumit	Bhopal	Python	70
DS1003	1003	vinit	Dhar	DBMS	90
DS1004	1004	Deepak	Ujjan	OR	60
DS1005	1005	Ajay	Indore	DBMS	80
DS1006	1006	Raj	Indore	Python	98

```
5 rows in set (0.00 sec)
```



```
mysql> savepoint s1;
```

Query OK, 0 rows affected (0.00 sec)


```
mysql> select * from student;
```

std_enroll_no	std_roll_no	std_name	address	subject	marks
DS1002	1002	Sumit	Bhopal	Python	70
DS1003	1003	vinit	Dhar	DBMS	90
DS1004	1004	Deepak	Ujjan	OR	60
DS1005	1005	Ajay	Indore	DBMS	80
DS1006	1006	Raj	Indore	Python	98

```
5 rows in set (0.00 sec)
```



```
mysql> savepoint s2;
```

Query OK, 0 rows affected (0.00 sec)


```
mysql> delete from student where std_enroll_no='DS1002';
```

Query OK, 1 row affected (0.00 sec)


```
mysql> delete from student where std_enroll_no='DS1003';
```

```
mysql> delete from student where std_enroll_no='DS1002';
Query OK, 1 row affected (0.00 sec)

mysql> delete from student where std_enroll_no='DS1003';
Query OK, 1 row affected (0.00 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1004        | 1004        | Deepak   | Ujjan   | OR       | 60    |
| DS1005        | 1005        | Ajay     | Indore  | DBMS     | 80    |
| DS1006        | 1006        | Raj      | Indore  | Python   | 98    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> rollback to s2;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+-----+-----+-----+-----+-----+-----+
| DS1002        | 1002        | Sumit    | Bhopal  | Python   | 70    |
| DS1003        | 1003        | vinit    | Dhar    | DBMS     | 90    |
| DS1004        | 1004        | Deepak   | Ujjan   | OR       | 60    |
| DS1005        | 1005        | Ajay     | Indore  | DBMS     | 80    |
| DS1006        | 1006        | Raj      | Indore  | Python   | 98    |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

- **Mysql> start transaction;**
- **Mysql> savepoint s1;**
- **Mysql> delete from student where student_enroll_no='DS1001';**
- **Mysql> savepoint s2;**
- **Mysql> rollback to s1;**